Cyber Security & Cryptography: Symmetric vs Asymmetric

DES, AES, RSA, Hash Functions, and Network Defense

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1. Information Security Triad: Confidentiality, Integrity, Availability CIA.
2. Classical Ciphers: Caesar, Playfair, Hill Cipher, Vigenere.
3. Symmetric Key Cryptography: DES structure and AES operation.
4. Asymmetric Key Cryptography: Diffie-Hellman Key Exchange and RSA Algorithm.
5. Message Authentication Codes MAC and SHA-256 Hashing.
6. Common Attacks: Phishing, SQL Injection, Cross-Site Scripting XSS, Man-in-the-Middle.